

## Build v32

# Prime-State All-Pass Geometry and the Observability Boundary

A rigorous attempt to turn identity, action, bidirectionality, and positivity into the missing Jordan shell estimate

### FINAL STATUS

The higher-dimensional prime object is constructed exactly. The 2-over-1 energy law is exact. The Ramanujan colors are an exact projected face of a global bidirectional unitary. The remaining cumulative observation map is unbounded, and controlling this one special all-prime state remains RH-equivalent. This build does not claim a proof of the Riemann Hypothesis.

The result is a genuine structural advance: it identifies where positivity lives, why it conserves energy, and why conservation alone cannot force the required cancellation.

# 1. Outcome first

The proposed picture survives translation into mathematics. A positive integer is a finite-excitation point of an infinite prime-exponent lattice. Multiplying by a prime is a shift along one coordinate. Extending the lattice in both directions makes every prime shift reversible. A convergent product of local all-pass factors then produces one global unitary state.



The first four arrows above are bounded and exact. The fifth arrow - the cumulative camera that converts periodic colors into long-range Jordan error - is unbounded. The RH-strength question is whether the one distinguished all-prime state nevertheless lies in that camera's domain.

## PLAIN-LANGUAGE READING

The engine is lossless; the camera is not uniformly well behaved. Knowing the engine has finite internal energy does not tell us that the image produced by an infinitely magnifying camera has finite energy. The theorem-sized task is to prove that this particular image remains controlled.

| CLAIM                                                                    | STATUS                  |
|--------------------------------------------------------------------------|-------------------------|
| Prime-exponent identity lattice and prime actions                        | PROVED                  |
| Global bilateral all-prime unitary $\Omega_a$ for $a > 1$                | PROVED                  |
| Ramanujan colors as the negative-squarefree face                         | PROVED                  |
| Literal one-state, two-channel conservation law                          | PROVED                  |
| Bounded cumulative synthesis on arbitrary color states                   | FALSE                   |
| Finite energy for the distinguished state along $\alpha_k \rightarrow 0$ | OPEN /<br>RH-EQUIVALENT |

## 2. Identity is the prime-exponent state

Every positive integer has one and only one prime-exponent vector. This is not an analogy; it is the fundamental theorem of arithmetic written as a state coordinate.

$$n \leftrightarrow (v_2(n), v_3(n), v_5(n), \dots), \\ F_+ = l_2(N) = \text{tensor}_p l_2(N_0), \quad e_n \leftrightarrow \text{tensor}_p e_{\{v_p(n)\}}.$$

Let  $S_p$  multiply a basis state by  $p$ , and let  $H$  measure logarithmic arithmetic height.

$$S_p e_n = e_{\{pn\}}, \quad H e_n = (\log n) e_n, \\ [H, S_p] = (\log p) S_p, \quad \text{Tr}(e^{\{-sH\}}) = \zeta(s) \quad \text{for } \text{Re}(s) > 1.$$

Thus identity and action are already joined: the state says which prime coordinates are occupied; the shift says how an added prime changes the state;  $H$  records the total logarithmic height. The partition sum of those heights is exactly  $\zeta$ .

For  $a=1+\alpha$  with  $0<\alpha<1$ , define  $r_p=p^{-a/2}$ . The finite squarefree Möbius state is

$$m_F = \prod_{\{p \in F\}} (I - r_p S_p) e_1 \\ = \sum_{\{d \text{ squarefree}, p|d \Rightarrow p \in F\}} \mu(d) d^{-a/2} e_d.$$

Because its squared coefficients sum to  $\zeta(a)/\zeta(2a)$ , the finite states converge in  $l_2$  to

$$m_a = \sum_{\{d \geq 1\}} \mu(d) d^{-a/2} e_d, \\ \|m_a\|^2 = \sum_{\{d \geq 1\}} \mu(d)^2 d^{-a} = \zeta(a)/\zeta(2a) < \infty.$$

### FIRST POSITIVITY

The coefficient state has a literal nonnegative norm. But that norm measures prime-coordinate amplitude, not cumulative arithmetic error. The distinction between those two energies is the central issue.

### 3. The cumulative camera and its exact fault line

Put  $K\alpha(u) = \sum_{m < u} m^{-\alpha} u^{\{1+\alpha\}} / (1+\alpha)$ . On the weighted physical space

$$\begin{aligned} K\text{-space}_\alpha &= L^2((0, \infty), x^{-\{a-1\}} dx), \\ (D_d F)(x) &= d^{\{a/2\}} F(x/d), \quad \|D_d F\| = \|F\|. \end{aligned}$$

Define the synthesis map on finitely supported states by  $W\alpha_e d = D_d K\alpha$ . It exactly intertwines prime action with physical dilation:

$$\begin{aligned} W\alpha S_p &= D_p W\alpha, \\ W\alpha m_F &= \sum_{\{d \text{ in the finite squarefree packet}\}} \mu(d) K\alpha(x/d). \end{aligned}$$

The formal infinite image is the extended Jordan error

$$\begin{aligned} W\alpha m_a &= E^{-\alpha}(x) = \sum_{\{d \geq 1\}} \mu(d) K\alpha(x/d), \\ \int_0^\infty E^{-\alpha}(x) x^{-\{s-1\}} dx &= \zeta(s-\alpha) / (s\zeta(s)). \end{aligned}$$

The physical norm is the Gram quadratic form

$$\begin{aligned} q\alpha(c) &= \|W\alpha c\|^2 \\ &= \sum_{\{m, n\}} c_m c_n^{-1} G\alpha(\log(m/n)). \end{aligned}$$

The normalized dilation cells are not a Bessel family. Equivalently,  $W\alpha$  is not bounded from the coefficient  $l_2$  space to the weighted  $K$ -space. Finite coefficient energy therefore cannot be transported through  $W\alpha$  by a generic norm inequality.

$$\begin{aligned} q\alpha((I - r_p S_p)c) &= (1 + r_p^2) q\alpha(c) \\ &\quad - 2r_p \operatorname{Re}(W\alpha c, D_p W\alpha c). \end{aligned}$$

#### THE MISSING SIGN

The Euler update contains one mixed correlation. Positivity makes the total a square, but it does not determine the sign or size of that cross term. The required prime/composite cancellation lives there.

## 4. Bilateral prime all-pass theorem

Extend every exponent coordinate from  $N_0$  to  $Z$ . The global space becomes  $l_2(Q_+)$ , and multiplication by a prime becomes a bilateral unitary  $U_p$  rather than a unilateral shift.

$$F_{bi} = \text{tensor}_p \{e_0\} \quad l_2(Z) = l_2(Q_+), \quad U_p e_r = e_{\{pr\}}, \\ \Omega_{\{p,a\}} = (I - r_p U_p^*) (I - r_p U_p)^{-1}.$$

Each  $\Omega_{\{p,a\}}$  is unitary. Its action on the local vacuum is

$$\Omega_{\{p,a\}} e_0 = -r_p e_{\{-1\}} \\ + (1 - r_p^2) \sum_{k \geq 0} r_p^k e_k.$$

This is the exact forward/backward prime cell: one backward excitation and the entire forward prime-power tail. Since  $\sum_p p^{-a} < \infty$  for  $a > 1$ , the infinite tensor product exists strongly:

$$\Omega_a = \text{tensor}_p \Omega_{\{p,a\}}, \quad \Omega_a^* \Omega_a = I.$$

For nested finite prime sets  $F \subset G$ , convergence follows from

$$\|\Omega_{Fe_1} - \Omega_{Ge_1}\|^2 \\ = 2[1 - \prod_{p \in G \setminus F} (1 - p^{-a})] \rightarrow 0.$$

The same Cauchy argument on elementary tensors and on the adjoints gives a global unitary, not merely a weak formal product.

### WHAT THE HIGHER DIMENSION IS

It is the bilateral prime-exponent lattice. The ordinary integers occupy its nonnegative cone. Reciprocal prime powers occupy the negative directions. The all-prime state joins both directions while preserving total Hilbert-space energy.

## 5. The colors are an exact face of that state

Project  $\Omega_a e_1$  onto states whose negative exponents are squarefree and whose other exponents are zero. For squarefree  $q$ , direct multiplication of the local amplitudes gives

$$\begin{aligned} & (e_{1/q}, \Omega_a e_1) \\ &= \prod_{p|q} (1 - p^{-a/2}) \prod_{p \nmid q} (1 - p^{-a}) \\ &= \mu(q) q^{a/2} / [\zeta(a) J_a(q)]. \end{aligned}$$

For nonsquarefree  $q$  the coefficient is zero, because a local vacuum image has no  $e_{-2}$ ,  $e_{-3}$ , ... component. That zero is exactly the Möbius squarefree rule.

Let  $c_q(n)$  be the Ramanujan sum and  $\phi_q(n) = c_q(n) / \sqrt{\phi(q)}$  its normalized Besicovitch  $B^2$  basis vector. On the negative-squarefree face define the diagonal contraction

$$D e_{1/q} = [\sqrt{\phi(q)} / q^{a/2}] \phi_q.$$

Then the projected state is precisely the normalized Jordan color function

$$\begin{aligned} D P_{-\Omega_a e_1} \\ &= \sum_{q \geq 1} \mu(q) c_q(n) / [\zeta(a) J_a(q)] \\ &= J_a(n) / n^a = j_a(n). \end{aligned}$$

So the colors do have a background equation: they are coordinates of one unitary all-prime state. Local  $q$ -colors and the global Jordan function are not two unrelated descriptions; they are a face and its recombination.

### SECOND POSITIVITY

The color coefficients have finite  $B^2$  energy because they arise through a contraction from a unit vector. The unresolved step is cumulative prefix observation, which amplifies long-wavelength  $q$ -colors and is not bounded on all  $B^2$  inputs.

## 6. The literal 2-over-1 energy cell

Set  $\beta=a/2$  and  $z=s-\beta$ . The all-pass scalar from Build v31 is

$$b_a(z)=(z-\beta)/(z+\beta)=(s-a)/s, \quad |b_a(it)|=1.$$

It has a one-state conservative realization with one input and one output:

$$\begin{aligned} x' &= -\beta x + \sqrt{a} u, \\ y &= u - \sqrt{a} x, \quad x(0)=-\alpha/\sqrt{a}. \end{aligned}$$

Its Laplace transform is exactly the arithmetic all-pass bridge

$$\begin{aligned} Y(z) &= b_a(z) \Phi \alpha(z+\beta) + \alpha/(z+\beta), \\ \text{equivalently } Y &= (s-a) \Phi \alpha(s)/s + \alpha/s. \end{aligned}$$

Most importantly, the storage balance is an identity:

$$\begin{aligned} d|x(t)|^2/dt &= |u(t)|^2 - |y(t)|^2, \\ |x(T)|^2 + \int_0^T |y|^2 &= |x(0)|^2 + \int_0^T |u|^2. \end{aligned}$$

This makes the user's '2 over 1' description exact: two boundary channels - incoming and outgoing - are coupled through one stored state. The initial  $\alpha/s$  term is stored reserve. No energy is created or destroyed.

### WHY THIS STILL DOES NOT FINISH RH

An all-pass system preserves finite energy; it does not turn infinite energy into finite energy. It transports the causality question without answering it. The remaining issue is whether the arithmetic input belongs to the one-sided finite-energy space in the first place.

## 7. The two-color cocycle

The low/high color split also has an exact state law. Write  $M = c\alpha x^a$  and, for  $0 < \lambda < 1$ ,

$$\begin{aligned} L\lambda &= \lambda E - (1-\lambda)M, & H\lambda &= (1-\lambda)E + (1-\lambda)M, \\ L\lambda + H\lambda &= E. \end{aligned}$$

For  $\lambda < \mu < 1$ , set  $r = (1-\mu)/(1-\lambda)$ . Then

$$\begin{aligned} [L\mu, H\mu]^T &= T_r [L\lambda, H\lambda]^T, \\ T_r &= \begin{bmatrix} 1 & 1-r \\ 0 & r \end{bmatrix}, & T_r T_s &= T_{rs}. \end{aligned}$$

The physical sum  $\Sigma = (I, I)$  obeys  $\Sigma T_r = \Sigma$ . The hidden anti-diagonal vector  $(-f, f)$  contracts by  $r$ , while the observable residual  $E$  has eigenvalue one.

The natural positive physical metric is singular:

$$G_\emptyset = \Sigma^* \Sigma = \begin{bmatrix} I & I \\ I & I \end{bmatrix}, \quad T_r^* G_\emptyset T_r = G_\emptyset.$$

Adding hidden storage with  $\kappa > 0$  gives an exact positive defect

$$\begin{aligned} G_\kappa &= \begin{bmatrix} I & I \\ I & (1+\kappa)I \end{bmatrix}, \\ (z, G_\kappa z) - (T_r z, G_\kappa T_r z) &= \kappa(1-r^2) \|H\|^2. \end{aligned}$$

That positivity controls only the hidden channel. It cannot shrink  $E$  because the cocycle preserves  $E$  exactly. There is no positive-definite invariant metric that makes a genuine eigenvalue  $r < 1$  an isometry.

### NO-GO THEOREM

The color cocycle can explain conservation and hidden contraction, but it cannot itself prove decay of the physical Jordan residual. A separate observability/correlation theorem is logically necessary.



## 8. What positivity actually says

For the true mixed shell state  $z_Y = (C_Y, \kappa_Y \alpha)^T$ , the physical observation is  $\Sigma z_Y = E \alpha$ . Its positive energy is

$$\begin{aligned} \|E \alpha\|_Y^2 &= (z_Y, G_0 z_Y) \\ &= \|C_Y + \kappa_Y \alpha\|_Y^2. \end{aligned}$$

Equivalently, it is twice the squared distance of  $z_Y$  from the cancellation plane  $\ker \Sigma$ . Expanding the square shows the exact required correlation:

$$\begin{aligned} &2 \operatorname{Re}(C_Y, \kappa_Y \alpha) \\ &= -\|C_Y\|^2 - \|\kappa_Y \alpha\|^2 + \text{small remainder}. \end{aligned}$$

The two component banks are individually large. To leave a small residual, they must approach antiparallel alignment with power-scale accuracy. Positivity certifies that the residual cannot be negative; it does not force the two banks to align.

### MOUNTAIN VERSION

The positive shell ledger is the total heat produced by every bump encountered over a band of the mountain. Each contribution is squared, so later bumps cannot erase earlier recorded heat. To keep total heat finite on an infinite descent, the body's exact shape and the terrain must make the disturbance on each farther band decay fast enough. Conservation tells us where the energy goes; arithmetic correlation determines whether the new heat becomes summable.

This is why 'infinite bumps' is the right objection. Local smoothness is irrelevant unless it comes with a uniform law across every scale. The gate is a statement about summability of the whole infinite sequence of shell energies.

## 9. Direct discrete attack: exact but theorem-sized

Let  $e\alpha(n) = \sum_{m \leq n} J\alpha(m) - c\alpha n^a$ . The sufficient dyadic target is

$$\begin{aligned} \sum_{Y < n \leq 2Y} |e\alpha(n)|^2 \\ << Y^{2+\varepsilon} + Y^{1+2\alpha+\varepsilon}, \\ \varepsilon < \min(\alpha, 1-\alpha). \end{aligned}$$

The exact hyperbola decomposition, valid for arbitrary  $D$  and  $H = \lfloor x/D \rfloor$ , is

$$\begin{aligned} S\alpha(x) = \sum_{d \leq D} \mu(d) P\alpha(x/d) \\ + \sum_{m \leq H} m^{\alpha M(x/m) - M(D)} P\alpha(H). \end{aligned}$$

Using the strongest standard Korobov-Vinogradov-shaped Mertens bound yields only

$$\begin{aligned} \sum_{Y < n \leq 2Y} |e\alpha(n)|^2 \\ << \alpha Y^3 \exp[-c\alpha(\log Y)^{3/5}(\log \log Y)^{-1/5}]. \end{aligned}$$

That is  $Y^{3-o(1)}$ . The target exponent  $\max(2, 1+2\alpha) + \varepsilon$  is strictly below 3 for  $0 < \alpha < 1$ . Current Möbius/hyperbola methods improve the size subexponentially but do not supply the fixed power saving.

Assuming the target for one  $\alpha$  analytically continues the relevant Dirichlet series past a shifted half-line and forces any denominator zero there to be paired with a numerator zero shifted by  $\alpha$ . Along distinct  $\alpha_k \rightarrow 0$ , those paired zeros would accumulate unless RH holds.

### DIAGNOSIS

The missing power saving is not routine technical slack. Along a determining null sequence, it is precisely strong enough to imply RH.

# 10. Farey and large-sieve attack

For a denominator block  $Q < q \leq R$ , put  $b_q = \mu(q)/[\zeta(a)J_a(q)]$  and  $\beta_{\{h/q\}} = b_q$ . The high-color tail has the exact positive Gram form

$$\int_Y^{2Y} |T_\alpha; Q, R(x)|^2 dx = \sum_{\{\theta, \varphi\}} \beta_\theta \beta_{-\varphi} K_\alpha(Y(\theta, \varphi)).$$

The matrix is positive semidefinite, but the desired smallness lies in its off-diagonal action on the one special Möbius vector  $\beta$ . Replacing  $\beta$  by  $|\beta|$  or estimating each denominator block separately destroys the cancellation.

The ordinary dual large sieve followed by the cumulative prefix matrix gives

$$\sum_{\{Y \leq k < 2Y\}} |\sum_{\{n \leq k\}} n^\alpha g_R(n)|^2 \ll Y^{2+2\alpha} (Y+R^2) R^{-2\alpha}.$$

At the Farey transition  $R \approx \sqrt{Y}$  this is  $Y^{3+\alpha}$ ; the target is  $Y^{2+\alpha}/(\log Y)^{1+\varepsilon}$ . The black-box route loses a complete factor  $Y$ , plus the required logarithm.

The full infinite tail can be regrouped causally with a weight  $0 \leq w_Q(d) \leq 1$ :

$$T_\alpha, Q(x) = \sum_{\{d \geq 1\}} \mu(d) w_Q(d) K_\alpha(x/d) = \sum_{\{dm < x\}} \mu(d) w_Q(d) m^\alpha.$$

This is an exact simplification, but its shell norm is still a signed Möbius-prefix quadratic form. No known uniform bilinear or large-sieve theorem controls it at the required strength.

## FAREY VERDICT

Spectrum-wide positivity is too coarse. The open information is the overlap of one arithmetic vector with the low-frequency eigenvectors after cumulative observation.

# 11. The exact surviving gate

Return to the raw Jordan summatory error

$$E\alpha(x) = \sum_{n \leq x} J\alpha(n) - x^{1+\alpha} / [(1+\alpha)\zeta(1+\alpha)].$$

Its positive dyadic shell ledger is

$$I\alpha(Y) = \int_Y^{2Y} |E\alpha(x)|^2 x^{-2-\alpha} dx, \\ \sum_{r \geq 0} I\alpha(2^r) < \infty.$$

Equivalently, after logarithmic unitary change of variables, the shells are orthogonal pieces of one  $L^2$  signal. Equivalently again, the Möbius all-prime state must lie in the domain of the closed cumulative synthesis associated with  $W\alpha$ .

For a determining sequence  $\alpha_k \downarrow 0$ , the following are equivalent formulations of the same RH-strength boundary:

| CLAIM                                                                  | STATUS      |
|------------------------------------------------------------------------|-------------|
| Jordan shell ledger is summable                                        | ENERGY FACE |
| Báez-Duarte fractional-part function has the weighted $L^2$ membership | MÖBIUS FACE |
| Burnol inverse Mellin state has the prescribed one-sided support       | CAUSAL FACE |
| Distinguished bilateral prime state survives cumulative observation    | STATE FACE  |

$$RH \Leftrightarrow \text{the gate holds for a determining sequence } \alpha_k \rightarrow 0^+.$$

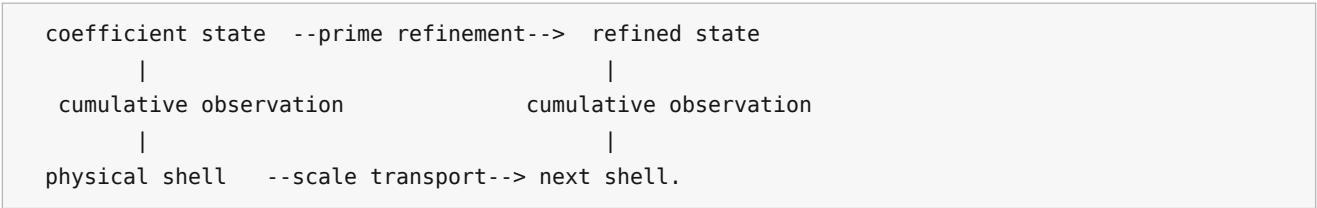
One fixed  $\alpha$  is insufficient because a zero in the denominator can in principle be canceled by a horizontally shifted numerator zero. A null sequence prevents persistent cancellation by discreteness of zeros.

# 12. The genuinely higher-dimensional formulation

The calculation identifies two independent axes that a complete argument must retain until the final square is taken.

| CLAIM                                                   | STATUS                |
|---------------------------------------------------------|-----------------------|
| Prime/color axis q or a finite prime set F              | IDENTITY / REFINEMENT |
| Logarithmic physical scale $u=\log x$ or dyadic shell Y | ACTION / OBSERVATION  |

Refining colors at fixed scale and moving to a new scale at fixed color cutoff do not commute harmlessly. Their mixed term is the arithmetic curvature of the two-parameter system. Collapsing either axis before preserving the signed cross terms reproduces one of the proved failures: finite Euler packets, blockwise Bessel estimates, or a coefficient-blind large sieve.



A viable theorem must control the defect of this square on the distinguished Möbius state. In operator language, it is a special-vector observability estimate. In arithmetic language, it is signed cancellation of the Euler-complete high-denominator tail. In the user's language, it is the one rule that simultaneously sees the ten-foot terrain and the entire mountain.

## NEW ATTACK SURFACE

Do not estimate colors independently. Do not square before Euler-complete cancellation. Seek a two-parameter Carleson/observability inequality that retains Möbius signs through both prime refinement and physical stopping.

# 13. Final theorem ledger

| CLAIM                                                                       | STATUS                |
|-----------------------------------------------------------------------------|-----------------------|
| Integers form the nonnegative cone of an infinite prime-exponent lattice    | PROVED                |
| The Möbius coefficients form a finite-energy state for every $a > 1$        | PROVED                |
| A global bilateral all-prime unitary $\Omega_a$ exists for every $a > 1$    | PROVED                |
| Its negative-squarefree face projects exactly to Ramanujan/Jordan colors    | PROVED                |
| The factor $(s-a)/s$ is a one-state, two-channel lossless system            | PROVED                |
| The low/high color split is an identity-preserving cocycle                  | PROVED                |
| Positivity forces decay of the observable Jordan residual                   | FALSE                 |
| Generic coefficient energy bounds cumulative shell energy                   | FALSE                 |
| Known Mertens/hyperbola estimates reach the needed fixed power saving       | FALSE                 |
| Known Farey/large-sieve estimates control the special vector after prefixes | FALSE                 |
| A determining sequence of shell memberships proves RH                       | PROVED<br>EQUIVALENCE |
| The determining sequence of memberships holds unconditionally               | OPEN                  |

## BOTTOM LINE

The proposed geometry did not collapse into metaphor. It became an exact unitary prime-state model and an exact conservation law. The attempt also proves why that geometry alone cannot finish RH: the final observation is unbounded, and positivity records the residual without forcing the necessary antiparallel alignment. The remaining problem is one special-vector, two-scale observability estimate.

That boundary is the mathematically honest stopping point of Build v32.

# References and provenance

The bilateral all-prime construction, its negative-face coefficient extraction, the exact 2/1 state realization, and their synthesis with the Prime DNA shell gate are derived in this build. Standard background and the previously identified equivalences are supported by the following primary or near-primary sources.

1. Báez-Duarte, L. A strengthening of the Nyman-Beurling criterion for the Riemann hypothesis. arXiv:math/0202141. <https://arxiv.org/pdf/math/0202141>
2. Burnol, J.-F. A note on Nyman's equivalent formulation of the Riemann hypothesis. arXiv:math/0202166. <https://arxiv.org/pdf/math/0202166>
3. Hedenmalm, H.; Lindqvist, P.; Seip, K. A Hilbert space of Dirichlet series and systems of dilated functions in  $L^2(0,1)$ . arXiv:math/9512211. <https://arxiv.org/pdf/math/9512211>
4. Lee, E.; Leong, S. New explicit bounds for the Mertens function. arXiv:2208.06141. <https://arxiv.org/abs/2208.06141>
5. Boca, F. P.; Radziwiłł, M. Limiting distribution of eigenvalues in the large sieve matrix. arXiv:1609.05843. <https://arxiv.org/abs/1609.05843>
6. Kaczorowski, J. On the mean square of the error term for an extended class of arithmetical functions. Monatsh. Math. (2013). <https://link.springer.com/article/10.1007/s00605-012-0417-6>
7. Ramanujan, S. On certain trigonometrical sums and their applications in the theory of numbers. Collected Papers, formula 9.6. <https://ramanujan.sirinudi.org/Volumes/published/ram21.html>
8. Suzuki, M. Hamiltonians arising from L-functions in the Selberg class. arXiv:1204.1827. <https://arxiv.org/abs/1204.1827>

## Notation note

Here  $0 < \alpha < 1$ ,  $a = 1 + \alpha$ ,  $c_\alpha = 1/[a\zeta(a)]$ ,  $J_\alpha$  is the Jordan totient of fractional order,  $\mu$  is the Möbius function,  $c_q$  is the Ramanujan sum, and  $\phi$  is Euler's totient. Endpoint conventions for integer cutoffs do not affect the  $L^2$  statements.

### CERTIFICATION

All displayed equalities labeled proved were checked algebraically against the Build v31 normalization. The RH-equivalent gate is stated as open. No proof square is attached to RH.